Rewarding Learning

ADVANCED SUBSIDIARY (AS)
General Certificate of Education
2026

Digital Technology

Assessment Unit AS 2

assessing

Fundamentals of Digital Technology

[SDT21]

TUESDAY 26 MAY, MORNING

MARK SCHEME

General Marking Instructions

Introduction

Mark schemes are published to assist teachers and students in their preparation for examinations. Through the mark schemes, teachers and students will be able to see what the examiners are looking for in response to questions and exactly where the marks have been awarded. The publishing of the mark schemes may help to show that examiners are not concerned about finding out what a student does not know but rather, with rewarding students for what they do know.

The purpose of mark schemes

Examination papers are set and revised by teams of examiners and revisers appointed by the Council. The teams of examiners and revisers include experienced teachers who are familiar with the level and standards expected of students in schools and colleges.

The job of the examiners is to set the questions and the mark schemes; and the job of the revisers is to review the questions and mark schemes commenting on a large range of issues about which they must be satisfied before the question papers and mark schemes are finalised.

The questions and the mark schemes are developed in association with each other so that the issues of differentiation and positive achievement can be addressed right from the start. Mark schemes, therefore, are regarded as part of an integral process which begins with the setting of questions and ends with the marking of the examination.

The main purpose of the mark scheme is to provide a uniform basis for the marking process so that all the markers are following exactly the same instructions and making the same judgements in so far as this is possible. Before marking begins, a standardising meeting is held where all the markers are briefed using the mark scheme and samples of the students' work in the form of scripts. Consideration is also given at this stage to any comments on the operational papers received from teachers and their organisations. During this meeting, and up to and including the end of the marking, there is provision for amendments to be made to the mark scheme. The document published represents the final form of the mark scheme.

It is important to recognise that in some cases there may well be other correct responses which are equally acceptable to those published: the mark scheme can only cover those responses which emerged in the examination. There may also be instances where certain judgements may have to be left to the experience of the examiner, for example where there is no absolute correct response – all teachers will be familiar with making such judgements.

Listing

Where one response is required to gain a mark, candidates will not gain credit if a correct response is given alongside one or more incorrect responses. This is referred to as listing.

		AVAILABLE MARKS
1	**(a)** Smallest value 0 [1] Largest value 15 [1]	
	(b) 1024 × 8 = 8192 bits [1] 8192 × 2 = 16384 bits [1]	
	(c) Binary conversion 11100_2 **or** Place value of 6th bit is 32 [1] 5 bits [1]	
	(d) 128 64 32 16 8 4 2 1 0 1 0 0 0 1 0 0 Correct conversion (either by place values or repeatedly dividing by 2) [1] 8 bits [1]	
	(e) +22 as an 8 bit binary number 00010110 [1] Invert bits 11101001 [1] Add 1 to LSB 11101010 [1]	
	(f) Place values/inversion [1] Answer −122 [1] for minus sign [1] for magnitude	14
2	**(a)** **(i)** **Data** Raw or unprocessed facts/figures [1] 65 is just a sequence of characters [1] **Information** Data processed/given a meaning [1] 65 is a student's test score [1]	
	(ii) **Knowledge** The result of applying rules to information [1] … to make decisions [1] Students S102 has failed and needs additional tuition [1]	
	(b) **4** is a hash total [1] Any **one** from: a count of the number of records transmitted [1] Has no other purpose than a count [1] **740** is a control total [1] Any **one** from: It also represents the total pay [1] Has an additional use or purpose [1]	
	(c) 5 7 2 4 9 Weight **6 5 4 3 2** [1] Product **30 35 8 12 18** [1] Sum **103** [1] Divide by 11 **9 r 4** [1] Subtract remainder **11 − 4 = 7** [1] MAX [4] for process Answer – – – – – – – – – 7 [1] 5 × [1]	

AVAILABLE MARKS

(d) Features of double entry

- Enter data twice
- Computer compares both values/automatic process
- Computer requests re-entry if values do not match/signals an error
- Example: User required to enter password twice when changing the password

Features of proof reading

- Visual checking
- Comparing data to the original source during entry process
- Data re-entered if both do not match
- Example: User compares what has been entered to the source form/ call data out to original source

Effectiveness of data verification in ensuring data entry is correct
Verification can guarantee integrity of data (BUT source must be correct)

Level	Marking criteria	Marks
Band 2	The candidate • Provides a detailed **description** of two types of data verification • Discusses the effectiveness of verification in ensuring that data entry is correct • Uses the appropriate Digital Technology terminology accurately throughout the response Presentation, spelling, punctuation and grammar are of a high standard.	[5]–[6]
Band 1	The candidate • Provides a detailed **description** of two types of data verification • Uses some relevant Digital Technology terminology Presentation, spelling, punctuation and grammar are sufficiently competent to make the response clear.	[3]–[4]
Band 0	The candidate • Provides a **description** of one/two methods of data verification which lack detail • Makes limited use of Digital Technology terminology Presentation, spelling, punctuation and grammar are such that the intended meaning is not completely clear.	[1]–[2]

[6]

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AVAILABLE MARKS

3 (a) (i) More/faster RAM [1]
… to support the fast processing speed/access (required for games) [1]
2 × [1]
More processors [1]
… to support the high processing power (required for games) [1]
2 × [1]
More cache memory [1]
… to enable fast retrieval of instructions/data (required for games) [1]
2 × [1]
Any 2 × [2]

(ii) RAM
RAM is needed to store instructions
… and data
… temporarily
… during execution
3 × [1]

Secondary storage
Secondary storage is needed to store software
… and data
… permanently
Secondary storage has the high capacity needed
Secondary storage allows data to be transferred to other computers/devices
3 × [1]

(iii) No moving parts
Higher speed access to data
Low power needed
Portable
Non-volatile
Can easily be rewritten
No need to defragment
3 × [1]

(b) Clock
Synchronises (all operations/calculations)
Measured by ticks/hertz/cycles per second
Measure of rate a computer can execute instructions
Ensures all circuits/components work together
2 × [1]

Cache memory
Used to improve processing speed
Stores frequently/recently used data/instructions from main memory
Memory closer to processor/faster access than main memory/Cache is searched first
2 × [1]

(c)

Description	Component
Stores data as it is fetched from memory	MDR/Memory Data Register [1]
Transfers commands from the CPU to other components	Control bus [1]
Is incremented to indicate the next instruction to be executed	PC/Program Counter [1]

(d) Features of multiprocessing

Two or more parts of a program/application (two or more programs) can be executed simultaneously
Two or more processors usually required working together
Each may handle a specific task
Alternatively, processing may be shared between processors for more complex tasks

Features of multitasking

Enables a user to perform more than one task at a time (apparently)
Only one task is running at any one time but it appears as though the tasks are running simultaneously
The processor automatically switches/allocates resources from one task to another

Use

Multiprocessing Example: Any large organisation-wide applications supporting complex tasks
Multitasking Example: Suitable, e.g. such as working on a spreadsheet application while running database reports

Level	Marking criteria	Marks
Band 2	The candidate • Provides a detailed and accurate description of both types of processing • Discusses how each type of processing could be used in a large organisation • Uses the appropriate Digital Technology terminology accurately throughout the response Presentation, spelling, punctuation and grammar are of a high standard.	[5]–[6]
Band 1	The candidate • Provides a detailed and accurate description of both types of processing • Uses some relevant Digital Technology terminology Presentation, spelling, punctuation and grammar are sufficiently competent to make the response clear.	[3]–[4]
Band 0	The candidate • Provides a description of both types of processing which is correct but lacks detail or provides a detailed description of one type of processing • Makes limited use of Digital Technology terminology Presentation, spelling, punctuation and grammar are such that the intended meaning is not completely clear.	[1]–[2]

[6]

AVAILABLE MARKS

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AVAILABLE MARKS

4 **(a)** Manage/coordinate hardware resources
… and software resources
File management
Process/task management
Provide the user interface
Implement system security
2 × [1]

(b) **File backup**
Creates a duplicate of data
… in case of corruption/data loss due to virus/accidental deletion
Backup copies are used to restore/recover files
May be cloud-based/use external storage devices
2 × [1]

Archive software
Refers to data which is no longer in regular/current use
Ensure the data is still available if needed
Data may be needed for legal/auditing/historical reasons
May be compressed
2 × [1]

(c) Enables the user to fill in empty fields in a window/panel
Resembles hard copy form
Controlled sequence of steps/provides prompts/instructions to help user
Examples of data entry tools
Provide validation
2 × [1]

(d) Real-time
Data is processed quickly to meet real world deadlines
Reacts to input data immediately
1 × [1]

Batch
Processing only takes place at intervals/when sufficient data is available
Delay in processing/output not available immediately
1 × [1]
Max [3] for descriptive comments

Justification
For safety reasons/efficiency
Real-time processing enables ATC to react immediately
Batch processing cannot respond quickly enough
2 × [1]
Max [2] for justification

Max [4] for combined descriptive/justification comments

(e) **(i)** User requirements fully met
No additional overheads due to unnecessary features
The development team available for troubleshooting/help/training [1]
2 × [1]

(ii) Immediate availability
Cost spread over many users
Existing user experiences available, e.g. user groups
2 × [1]

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AVAILABLE MARKS

5 (a)

Part of domain name	Letter
www.ccea.org.uk	**C** Identifies the domain name [1]
.org	**B** Top level domain [1]
ccea.org	**A** Second level domain [1]
uk	**D** Country level domain [1]

(b) The domain name is used instead of the IP address/text form of IP address [1]
(www.)clarendonhigh.ac is the domain name [1]

The URL is the complete address of a web resource [1]
http://www.clarendonhigh.ac.uk/digitaltechnology (takes the user to the digital technology page on the website) [1]

(c) Change 1
Line 1 or **<h2>Topics</h2>** [1]
… becomes **<h1>Topics</h1>** [1]

Change 2
Line 2 or **<ul>** becomes **<ol>** [1]
Line 6 or **
** becomes **
** [1]

(d) Note: 'how the intranet could be used by teachers'

Features of an intranet [0]

Learning resources
Teachers cooperating [1]
…..with example….such as develop schemes [1]
Produce materials [1]
……with example….such as develop a learning environment for class [1]
Distributing materials [1]
…..with example…..such as digital books/video/hyperlinks [1]
2 × [1]

Communication
Sending messages/communicating [1]
with example…..such as…to parents/pupils [1]
Providing information [1]
with example….such as displaying timetables [1]
Organising event [1]
…..with example….such as assessment/prize day/sports day/meetings [1]
2 × [1]

2 × [2]

	AVAILABLE MARKS

(e) **Client side**

When a pupil is carrying out a transaction over the internet, processing is done on the user's computer
May use scripting to support interaction on web page
Code needed is downloaded and executed by browser or a plugin
e.g. to ensure name and class entered/ensure items selected/give feedback if section not completed

Server side

After client-side processing
Data sent across internet to server
Server runs an application to process the information and returns results/server completes processing
May use PHP
e.g. process payment and send email confirmation/return message if item out of stock

Level	Marking criteria	Marks
Band 2	The candidate • Provides a full description of both client-side and server-side processing • The response includes examples for both from the scenario provided • Uses the appropriate Digital Technology terminology accurately throughout the response Presentation, spelling, punctuation and grammar are of a high standard.	[5]–[6]
Band 1	The candidate • Provides a detailed description of both client-side and server-side processing which is accurate • Uses some relevant Digital Technology terminology Presentation, spelling, punctuation and grammar are sufficiently competent to make the response clear.	[3]–[4]
Band 0	The candidate • Provides a description of both client-side and server-side processing or provides a detailed description of either client-side or server-side processing • Makes limited use of Digital Technology terminology Presentation, spelling, punctuation and grammar are such that the intended meaning is not completely clear.	[1]–[2]

[6] 22

Total **100**